PUPIL: _____ CLASS: _____ TEACHER: _____

KEY TOPIC 5 • 6 ENQUIRY HOURS • 1914–1918

The British Sector of the Western Front, 1914–1918

Injuries, Treatment and the Trenches in the British Sector



Primary Visual Evidence: *Stretcher Bearers of the RAMC Lifting a Wounded Man out of a Trench* • Gilbert Rogers (c. 1919)

OFFICIAL EDEXCEL SOURCE ARCHIVE • IWM ART 2485 / RAMC HISTORIC COLLECTION

1. FEATURE DRILLS [4M]
Q1(a) & Q1(b): Two 2-mark questions. Identify 1 specific feature + 1 supporting factual detail (5 mins).

2. SOURCE UTILITY [8M]
Q2(a): 3-step evaluation of Content, Provenance (Nature, Origin, Motive), and Contextual Limitations (10 mins).

3. FOLLOW-UP ENQUIRY [4M]
Q2(b): 4-row tabular enquiry: Detail, Historical Question, Source Type, and Clinical Value (5 mins).

COURSE SPECIFICATION • KEY ENQUIRY SEQUENCE (1914–1918)	LEARNT	REVISED
Key Topic 5.1: The Theatre of War: The British Sector, Trench Geography & Battles *Why were trenches designed in zig-zag traverses, and how did terrain at Ypres, Somme, and Arras affect medical treatment?*	☐	☐
Key Topic 5.2: The Trench Environment: Mud, Vermin & Non-Combat Illnesses *How did frontline conditions cause trench foot, trench fever, and dysentery, and what preventive routines were enforced?*	☐	☐
Key Topic 5.3: Battlefield Trauma: High Explosive Shrapnel, Gas Attacks & Infection *Why did explosive artillery produce gas gangrene, and how did the Brodie steel helmet and respirators reduce fatalities?*	☐	☐
Key Topic 5.4: The Chain of Evacuation: Stretcher Bearers, RAP, Dressing Stations & CCS *How did the RAMC and FANY evacuate wounded soldiers through six stations, and why was triage critical at CCSs?*	☐	☐
Key Topic 5.5: Surgical Breakthroughs: The Thomas Splint, Wound Debridement & Mobile X-Rays *How did the Thomas Splint slash femur mortality from 80% to 20%, and why was Carrel-Dakin irrigation necessary?*	☐	☐
Key Topic 5.6: Lifesaving Innovations: Blood Storage, Brain Surgery & Plastic Reconstruction *How did Oswald Robertson establish the first blood bank at Cambrai, and how did Gillies rebuild shattered faces at Sidcup?*	☐	☐

LIVING TIMELINE • PART 1: FRONTLINE CONDITIONS & EARLY TRAUMA (1914–1915)

Instructions: As you study each enquiry lesson, complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

OCT 1914 • Flanders Mud & The 1st Battle of Ypres

Key Topic 5.1

The British Expeditionary Force (BEF) holds the Ypres Salient. Low-lying Flanders clay creates a dangerously high water table; trenches fill with liquid mud. Shelling smashes all natural drainage canals, leaving men standing waist-deep in water and giving rise to the first epidemics of trench foot and dysentery.

APR 1915 • 2nd Battle of Ypres: The First Poison Gas Attack

Key Topic 5.2 • 5.3

German forces release 160 tons of chlorine gas across a 4-mile front. British troops suffer suffocation and fluid in the lungs; soldiers improvise with urine-soaked pads before chemically impregnated cotton-pad respirators (the Hypo Helmet) are rushed to the front. Lice-borne trench fever breaks out across all divisions.

DEC 1915 • The Thomas Splint & Introduction of the Brodie Helmet

Key Topic 5.3 • 5.5

Robert Jones introduces the Thomas Splint to frontline medical posts, stabilizing compound fractures and reducing femur mortality from 80% to 20%. Simultaneously, the War Office begins issuing John Brodie's stamped steel helmet, reducing penetrating shrapnel head trauma by an estimated 75–80%.

If you think your walk to period 1 is rough, try carrying a 14-stone sergeant through 4 miles of knee-deep Flanders clay.

LIVING TIMELINE • PART 2: MASS OFFENSIVES & CLINICAL INNOVATION (1916–1917)

Instructions: Complete the timeline missions by sketching and annotating in the corresponding Key Topic boxes below.

JULY 1916 • The Battle of the Somme: The Triage Crisis

Key Topic 5.4

The BEF suffers 57,000 casualties on 1 July 1916. Stretcher bearers and Casualty Clearing Stations are overwhelmed; motor ambulances struggle along single rutted roads. As a direct result, the RAMC enforces a strict three-tier triage system (Walking Wounded, Immediate Surgery, Moribund) to prevent total collapse.

APR–NOV 1917 • Arras Underground Hospital & Passchendaele Mud

Key Topic 5.1 • 5.4

At Arras, New Zealand tunnelers excavate Thompson's Cave—a 700-bed underground hospital with operating theatres, running water, and electric lights 20m beneath the chalk. At 3rd Ypres (Passchendaele), artillery destroys all drainage systems, creating knee-deep liquid mud requiring 6 bearers per stretcher.

NOV 1917 • Oswald Robertson's Blood Depot & Gillies Facial Reconstruction

Key Topic 5.6

At the Battle of Cambrai, Captain Oswald Robertson deploys the world's first blood bank, using sodium citrate to stop clotting, glucose to preserve red cells, and ice chests to store Group O blood for 28 days. In Britain, Harold Gillies pioneers plastic facial surgery using tubed pedicles at Queen's Hospital, Sidcup.

How did the physical terrain and trench architecture of the British sector dictate medical evacuation and care?

Core Specification Focus: The British sector of the Western Front: theatre of war in Flanders and northern France; Ypres, Somme, Arras, Cambrai; trench system (frontline, support, reserve, communication trenches; saps); terrain, clay, water table, mud; underground systems at Arras (Thompson's Cave).

CHRONOLOGICAL INQUIRY SPINE • 5 CAUSAL MILESTONES

PUPIL NOTE-TAKING ZONE • *Construct concise causal notes alongside each milestone prompt*

1 1914–1915 · Why was the low-lying topography of Ypres a tactical and medical nightmare? **[5.1.1]**	
• **Digging into sticky wet clay**	
• **Heavy artillery breaking water canals**	
• **Wounded men sinking into knee-deep mud**	

2 1914–1918 · How did zig-zag traverses limit shrapnel injuries and contain shell explosions? **[5.1.2]**	
• **Digging sharp zig-zag corners into earth**	
• **Laying wooden duckboards over smelly puddles**	
• **Stacking sandbags high on front firing steps**	

3 1915–1918 · Why did narrow communication trenches create acute bottlenecks for stretchers? **[5.1.3]**	
• **Guarding firing steps facing No Man's Land**	
• **Squeezing wooden stretchers along narrow passages**	
• **Crawling through listening saps near enemy barbed wire**	

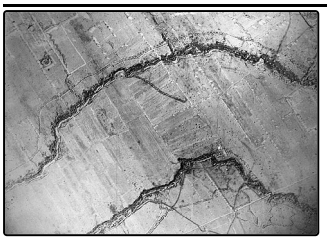
4 Nov 1916–1917 · How did the chalk geology at Arras allow safe, sterile surgical conditions? **[5.1.4]**	
• **Carving hospital wards deep into dry white chalk**	
• **Wiring electric light bulbs and water pipes**	
• **Treating seven hundred beds safely underground**	

5 1916–1917 · Why did mass-casualty offensives force the RAMC to reorganize evacuation? **[5.1.5]**	
• **Fifty-seven thousand men falling on first morning**	
• **Driving heavy metal tanks over chalk slopes**	
• **Overworked medical orderly lines facing endless stretchers**	

Welcome to Flanders: 500 square miles of fermented pig manure, liquid mud, and artillery with terrifying accuracy.

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

SOURCE A • CONTEMPORARY AERIAL RECONNAISSANCE • ROYAL FLYING CORPS **ORIGIN: 1917**



Royal Flying Corps aerial photograph showing the intricate zig-zag traverses of frontline, support, and communication trenches near Ypres, criss-crossed with artillery shell craters.

Historical Provenance: Official military aerial photograph taken by the Royal Flying Corps, British Sector near Ypres, 1917. Imperial War Museum Collection.

Study Source A. How useful is Source A for an enquiry into the defensive layout and tactical design of trench systems on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an official military aerial reconnaissance photograph taken by the Royal Flying Corps in 1917. What does an overhead view reveal about traverses, fire-steps, and saps, and what ground-level medical and mud conditions does it omit?

ACCESSIBLE WRITING FRAMES:

Source A is useful for an enquiry into this topic because... • Specifically, the source reveals... • In terms of provenance, considering the author and date... • However, the utility is limited because... • Therefore...

KEY VOCABULARY BANK:

Royal Flying Corps • zig-zag traverses • communication trench • Ypres Salient • Messines Ridge • water table • duckboards • parapet • parados • fire-step

Write your response to Question 2(a) in the space provided below:

(Response continues on facing page →)

TIMELINE MISSION Turn to Pages 2–3 (Milestone 1). Sketch and annotate the zig-zag traverse pattern and low-lying Ypres water table. **[ ] Done**

Trench drainage tip: If your duckboards are floating, you are no longer in an infantry trench; you are commanding a submarine.

Continue your response to Question 2(a) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(a):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the system of communication trenches on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the underground hospital at Arras (Thompson's Cave). **[/2]**

FEATURE:

DETAIL:

Why were environmental pathogens and vermin as dangerous to British troops as enemy artillery?

Core Specification Focus: Ill health arising from the trench environment: trench foot (pathology, gangrene, prevention, whale oil, sock inspection); trench fever (body lice, Pediculus humanus, delousing stations, Serbian barrels); shell shock (symptoms, NYDN, rest treatment); dysentery (water chlorination, chloride of lime, latrines); underground shelter conditions.

CHRONOLOGICAL INQUIRY SPINE • 5 CAUSAL MILESTONES

PUPIL NOTE-TAKING ZONE • *Construct concise causal notes alongside each milestone prompt*

1 1914–1915 · How did prolonged cold water immersion destroy tissue and produce gangrene? [5.2.1]

- **Standing for days in freezing dirty water**
- **Toes turning numb, swollen, and bluish-black**
- **Surgeons sawing off rotting gangrene feet**

2 1915–1918 · Why did mandatory officer inspections succeed where voluntary advice had failed? [5.2.2]

- **Pairs of soldiers rubbing fishy grease into toes**
- **Packing three dry woollen pairs in kits**
- **Officers holding daily morning foot inspections**

3 1914–1918 · Why did trench fever persist until the body louse insect vector was proven in 1918? [5.2.3]

- **Scratching itchy insect bites along uniform seams**
- **High shaking chills and burning shin bone aches**
- **Steaming dirty woollen jackets in giant barrels**

4 1914–1918 · How did strict sanitation and chemical water chlorination prevent epidemics? [5.2.4]

- **Violent stomach cramps from dirty swamp water**
- **Digging deep toilet pits far behind sleeping lines**
- **Throwing smelly white lime powder into waste trenches**

5 1914–1918 · Why was shell shock initially stigmatised as cowardice before being treated medically? [5.2.5]

- **Endless artillery thunder breaking soldiers' minds**
- **Men shaking uncontrollably and losing their speech**
- **Resting traumatised patients in quiet country hospitals**

Whale oil smells like dead fish and regrets, but it beats having your toes amputated by an RAMC surgeon with a bone-saw.

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT **ORIGIN: July 1916**



British infantrymen of 'A' Company, 11th Battalion, Cheshire Regiment, resting wrapped in waterproof groundsheets on the duckboards and dugouts of a captured trench at Ovillers-la-Boisselle, while a sentry remains on duty on the parapet.

Historical Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke during the Battle of the Somme, July 1916. Imperial War Museum Collection.

Study Source A. How could you follow up Source A to find out more about the living conditions and non-combat illnesses experienced by British soldiers in frontline trenches on the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

ACCESSIBLE WRITING FRAMES:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

trench foot • whale oil • buddy-system • gangrene • trench fever • body lice (Pediculus humanus) • delousing stations • Serbian barrels • shell shock • NYDN • dysentery • chloride of lime

Complete the 4-prompt historical enquiry matrix below:

(Response continues on facing page →)

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

Disciplinary Commentary • Clinical Justification of Chosen Source Type:

TIMELINE MISSION Turn to Pages 2–3 (Milestone 2). Draw soldiers applying whale oil and delousing garments in steam ovens. **[ ] Done**

Continue your response to Question 2(b) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(b):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the methods used to prevent trench foot on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the causes of trench fever among British troops. **[/2]**

FEATURE:

DETAIL:

How did high-explosive artillery trauma and chemical poison gas transform surgical treatment?

Core Specification Focus: Wounds, injuries and diseases: high-explosive artillery shells, shrapnel fragments; infection (gas gangrene, Clostridium welchii, tetanus); head trauma and the Brodie helmet; chemical gas attacks (chlorine, phosgene, mustard gas) and respirators.

CHRONOLOGICAL INQUIRY SPINE • 5 CAUSAL MILESTONES

PUPIL NOTE-TAKING ZONE • *Construct concise causal notes alongside each milestone prompt*

1 1914–1918 • Why did jagged shrapnel fragments cause far worse infections than rifle bullets? [5.3.1]

• **Blasting hot jagged metal balls through flesh**	
• **Blowing dirty uniform cloth deep into broken bones**	
• **Causing over half of all battlefield wounds**	

2 1914–1918 • Why was Clostridium welchii so lethal in deep, airless shrapnel wounds? [5.3.2]

• **Fertilising farmland with rotten animal manure**	
• **Deadly microbes multiplying inside airless wounds**	
• **Hissing bubbles swelling under blackened skin**	

3 Late 1915 • Why did the introduction of the Brodie helmet cause an apparent rise in hospital head cases? [5.3.3]

• **Soft cloth caps offering zero protection from blasts**	
• **Pressing hard steel bowls with wide gutters**	
• **Slashing deadly skull splinter wounds by eighty percent**	

4 Apr 1915–1916 • How did the rapid evolution of chemical respirators counter suffocating gases? [5.3.4]

• **Releasing green-yellow choking cloud over fields**	
• **Holding urine-soaked handkerchiefs over noses**	
• **Breathing through grey flannel hoods with glass eyepieces**	

5 1916–1918 • Why was mustard gas far harder to counter than chlorine or phosgene? [5.3.5]

• **Sucking air through rubber tubes and charcoal canisters**	
• **Oily yellow liquid burning huge skin blisters through clothes**	
• **Soaking into muddy earth for weeks without evaporating**	

The Brodie helmet: Looks like an upside-down soup bowl, but prevents your skull from becoming one.

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

SOURCE B • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • MACHINE GUN CORPS **ORIGIN: July 1916**



A British Machine Gun Corps crew operating a Vickers machine gun from an improvised trench position near Ovillers on the Somme while wearing issue Phenate-Hexamine (PH) anti-gas helmets with exhale valves.

Historical Provenance: Official British War Office photograph taken by Lieutenant John Warwick Brooke near Ovillers during the Battle of the Somme, July 1916. Imperial War Museum Collection.

Study Source B. How useful is Source B for an enquiry into the methods used to protect British soldiers from poison gas attacks on the Western Front? Explain your answer, using Source B and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic British War Office Phenate-Hexamine (PH) anti-gas helmet manufactured in 1915–1916. What physical protective properties does chemically impregnated flannel possess, and what severe combat limitations did it have in battle?

ACCESSIBLE WRITING FRAMES:

Source B is useful for investigating this topic because... • Specifically, the source provides evidence that... • In terms of provenance, considering the nature and purpose... • However, the source is limited because... • Consequently...

KEY VOCABULARY BANK:

high-explosive artillery • shrapnel balls • gas gangrene • Clostridium welchii • Brodie steel helmet • chlorine gas (1915) • phosgene • mustard gas (1917) • PH helmet • Small Box Respirator (1916)

Write your response to Question 2(a) in the space provided below:

(Response continues on facing page →)

TIMELINE MISSION Turn to Pages 2–3 (Milestone 2). Sketch the Brodie steel helmet and British PH gas hood. **[ ] Done**

Continue your response to Question 2(a) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(a):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the effects of poison gas attacks on soldiers on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the design of the British Brodie steel helmet. **[/2]**

FEATURE:

DETAIL:

How successfully did the Chain of Evacuation transport, triage, and treat mass battlefield casualties?

Core Specification Focus: The work of the RAMC and FANY: the Chain of Evacuation (stretcher bearers, Regimental Aid Posts, Field Ambulances and Dressing Stations, Casualty Clearing Stations, Base Hospitals); triage systems; transport methods (motor ambulances, ambulance trains, canal barges).

CHRONOLOGICAL INQUIRY SPINE • 5 CAUSAL MILESTONES

PUPIL NOTE-TAKING ZONE • *Construct concise causal notes alongside each milestone prompt*

1 1914–1918 • Why was rapid evacuation by stretcher bearers vital to prevent irreversible shock? [5.4.1]	
• Teams of four hauling wounded through bullet fire	
• Crowding into dark sandbagged frontline dugouts	
• Slapping brown iodine and white bandages on cuts	

2 1914–1918 • What was the tactical role of Dressing Stations in preventing CCS congestion? [5.4.2]	
• Setting up treatment cots inside ruined farm barns	
• Checking clean bandages one mile behind frontline	
• Loading urgent bleeding cases into transport carts	

3 1914–1918 • How did the 3-tier triage system maximize survival during mass offensives? [5.4.3]	
• Setting up big operating tents near railway tracks	
• Sorting bleeding soldiers into three survival groups	
• Rushing torn stomach wounds straight to surgeons' tables	

4 1914–1918 • Why were Base Hospitals situated near major coastal rail terminals and ports? [5.4.4]	
• Converting coastal resort hotels into recovery wards	
• Treating specialized amputations and poison burns	
• Carrying healed patients onto white hospital ships	

5 1914–1918 • Why were canal barges preferred over trains for head and chest trauma? [5.4.5]	
• Bouncing painfully in old wooden horse wagons	
• Donating hundreds of fast petrol motor vans	
• Gliding gently down quiet canals for head injuries	

The Regimental Aid Post: 200 yards from the German front line, lit by candle-ends, and smelling entirely of iodine and panic.

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

SOURCE A • CONTEMPORARY PHOTOGRAPHIC EVIDENCE • WESTERN FRONT **ORIGIN: August 1917**



A team of six British stretcher bearers struggling through deep, liquid Flanders mud to carry a wounded soldier on a wooden stretcher near Boesinghe during the Third Battle of Ypres.

Historical Provenance: Official British Army photograph taken by Lieutenant John Warwick Brooke near Boesinghe during the Third Battle of Ypres, 1 August 1917. Imperial War Museum Collection.

Study Source A. How could you follow up Source A to find out more about the difficulties stretcher bearers faced evacuating casualties from the Western Front? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

ACCESSIBLE WRITING FRAMES:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

RAMC • FANY • Regimental Aid Post • Advanced Dressing Station • Casualty Clearing Station • Base Hospital • triage system • motor ambulance • hospital train • canal barge

Complete the 4-prompt historical enquiry matrix below:

(Response continues on facing page →)

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

Disciplinary Commentary • Clinical Justification of Chosen Source Type:

TIMELINE MISSION Turn to Pages 2–3 (Milestone 5). Draw the 6 sequential stages of the Chain of Evacuation. **[ ] Done**

Continue your response to Question 2(b) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(b):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the triage system used at Casualty Clearing Stations on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the work of the First Aid Nursing Yeomanry (FANY) on the Western Front. **[/2]**

FEATURE:

DETAIL:

Why were the Thomas Splint, debridement, and mobile X-rays decisive surgical breakthroughs?

Core Specification Focus: Medical advances on the Western Front: the Thomas Splint (Hugh Owen Thomas, Robert Jones) reducing compound femur mortality; wound debridement and delayed primary closure; the Carrel-Dakin antiseptic irrigation method; mobile X-ray units and radiology.

CHRONOLOGICAL INQUIRY SPINE • 5 CAUSAL MILESTONES

PUPIL NOTE-TAKING ZONE • *Construct concise causal notes alongside each milestone prompt*

1 1914–1918 • Why was surgical excision of dead tissue more effective than chemical antiseptics? [5.5.1]	
• Cutting away bruised, black, and dead muscle tissue	
• Depriving hungry bacteria of rotting food	
• Leaving surgical cuts open to wash before stitching	

2 1915–1918 • How did continuous chemical irrigation solve the problem of deep wound infection? [5.5.2]	
• Pushing narrow rubber tubes deep into torn flesh	
• Dripping fresh salty bleach solution day and night	
• Killing hidden microbes before gangrene takes hold	

3 Dec 1915–1916 • How did rigid traction prevent fatal haemorrhage and shock from broken femurs? [5.5.3]	
• Fitting padded metal oval ring around upper groin	
• Pulling leather cords to stretch broken leg bones taut	
• Dropping thigh fracture deaths from eighty to twenty percent	

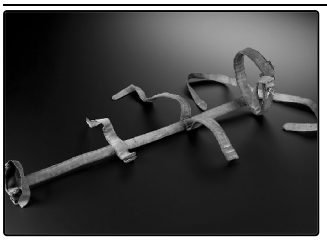
4 1914–1918 • Why was locating shrapnel on an X-ray vital before operating on gas gangrene? [5.5.4]	
• Running small dynamo generators off motor engines	
• Scanning wounded soldiers right behind battle lines	
• Locating buried steel balls before surgeon cuts	

5 1914–1918 • Why did undetected cloth fragments remain the primary cause of post-op gangrene? [5.5.5]	
• Shattering delicate glass vacuum bulbs on bumpy tracks	
• Overheating metal tubes after five minutes of use	
• Failing to spot buried uniform cloth and muddy splinters	

The Thomas Splint: Before Robert Jones introduced it, an 80% chance of death; after Jones, an 80% chance of living to complain about the food.

Question 2(a): Source Utility Assessment [8 marks • 10 mins]

SOURCE A • CONTEMPORARY MEDICAL ARTIFACT • STANDARD ISSUE **ORIGIN: c.1916**



An authentic surviving British Army Thomas Splint, showing the padded oval ring fitted against the groin, rigid metal side bars, and canvas support straps used to maintain traction and immobilize broken thigh bones.

Historical Provenance: Original First World War Thomas Splint preserved in the Wellcome Historical Medical Collection, London. Designed by Hugh Owen Thomas and introduced to the Western Front in 1915–1916 by Robert Jones.

Study Source A. How useful is Source A for an enquiry into the methods used to treat wounded soldiers with fractured limbs on the Western Front? Explain your answer, using Source A and your knowledge of the historical context. [8 marks]

Provenance Clue: Consider the nature of an authentic instructional photograph taken in an RAMC military hospital training ward c.1916. What does it demonstrate about mechanical traction and splinting, and why might an instructional training photograph differ from emergency frontline conditions in a mud dugout?

ACCESSIBLE WRITING FRAMES:

Source A is useful for an enquiry into this topic because... • Specifically, the source demonstrates that... • In terms of provenance, considering the contemporary context... • However, the utility is limited because... • Therefore...

KEY VOCABULARY BANK:

Thomas Splint • Hugh Owen Thomas • Robert Jones • compound fracture • femoral artery • mechanical traction • wound debridement • Carrel-Dakin solution • delayed primary closure • mobile X-ray unit

Write your response to Question 2(a) in the space provided below:

(Response continues on facing page →)

TIMELINE MISSION Turn to Pages 2–3 (Milestone 3). Sketch the Thomas Splint groin ring and traction cord. **[ ] Done**

Carrel-Dakin solution: If it burns like liquid fire and smells like a Victorian washhouse, congratulations—it is killing the gas gangrene.

Continue your response to Question 2(a) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(a):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of the Thomas Splint used on the Western Front. **[/2]**

FEATURE: _____

DETAIL: _____

1(b) Describe one feature of the Carrel-Dakin method of treating infected wounds. **[/2]**

FEATURE: _____

DETAIL: _____

How did stored blood banks, specialized neurosurgery, and plastic reconstruction transform wartime survival?

Core Specification Focus: Medical advances on the Western Front: blood transfusions and storage (Landsteiner, Hustin, Rous and Turner, Captain Oswald Robertson and the Cambrai blood bank); specialized neurosurgery (Harvey Cushing); plastic and facial reconstruction (Harold Gillies, Queen's Hospital Sidcup, tubed pedicle).

CHRONOLOGICAL INQUIRY SPINE • 5 CAUSAL MILESTONES

PUPIL NOTE-TAKING ZONE • *Construct concise causal notes alongside each milestone prompt*

1 1914–1915 • How did sodium citrate liberate blood transfusion from donor-to-patient tubing? [5.6.1]	
• **Adding chemical salts to stop blood turning into jelly**	
• **Uncoupling donor arm from dying patient's vein**	
• **Pouring red liquid freely through sterile glass funnels**	

2 1916–1917 • Why was glucose addition essential to prevent red blood cell breakdown? [5.6.2]	
• **Mixing sweet sugar syrup to feed fragile red cells**	
• **Packing glass medicine bottles into insulated ice chests**	
• **Keeping precious donations fresh for four weeks**	

3 Nov 1917 • Why did pre-stored blood transform surgical survival during surprise assaults? [5.6.3]	
• **Collecting universal Type O bottles before big attack**	
• **Carrying twenty-two vacuum metal flasks to frontline**	
• **Reviving dying soldiers suffering cold shock**	

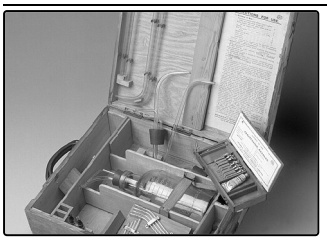
4 1917–1918 • Why was local anaesthesia superior to general anaesthetic in head trauma? [5.6.4]	
• **Numbing scalp with local needle jabs to stop brain swelling**	
• **Clamping bleeding blood vessels with tiny silver clips**	
• **Pulling out buried iron shards with magnetic rods**	

5 1917–1918 • How did the tubed pedicle technique prevent skin grafts from rotting and dying? [5.6.5]	
• **Rolling chest skin into living round meat tubes**	
• **Keeping blood pumping into graft while it stitches to face**	
• **Sculpting new noses, chins, and jaws for shattered soldiers**	

Robertson's Blood Depot: Ice chests, sodium citrate, and refrigerated blood at Cambrai—proof that cold beer isn't the only thing worth chilling.

Question 2(b): Follow-Up Enquiry Table [4 marks • 5 mins]

SOURCE A • PRIMARY MEDICAL ARTIFACT • CAMBRAI BLOOD DEPOT **ORIGIN: 1917**



The authentic wooden ice chest, glass collection flasks, rubber delivery tubing, and sodium citrate solutions utilized by Captain Oswald Robertson to establish the world's first blood bank during the Battle of Cambrai in November 1917.

Historical Provenance: Original portable blood transfusion outfit preserved in the Royal Army Medical Corps Historical Collection, London. Utilized during the Battle of Cambrai, November 1917.

Study Source A. How could you follow up Source A to find out more about the success of stored blood transfusions at the Battle of Cambrai in 1917? [4 marks]

Examiner Strategy: Follow up an exact quotation or visual detail. Your question must directly target your chosen detail. Your named source type must be authentic and specific (e.g. RAMC casualty returns, battalion war diaries).

ACCESSIBLE WRITING FRAMES:

A key detail to follow up is... • The question I would ask is... • The specific historical source type needed is... • This source would help answer my question because...

KEY VOCABULARY BANK:

Karl Landsteiner • ABO blood groups • sodium citrate (Hustin 1914) • glucose preservative (Rous & Turner 1916) • Oswald Robertson • Battle of Cambrai (1917) • blood depot • Harvey Cushing • silver clips & surgical magnets • Harold Gillies • Queen's Hospital Sidcup • tubed pedicle graft

Complete the 4-prompt historical enquiry matrix below:

(Response continues on facing page →)

Detail in Source A that I would follow up:	
Question I would ask:	
Type of source I would use:	
How this would help me find out more:	

Disciplinary Commentary • Clinical Justification of Chosen Source Type:

TIMELINE MISSION Turn to Pages 2–3 (Milestone 6). Sketch Oswald Robertson's iced blood chest and Harold Gillies' tubed pedicle. **[ ] Done**

QUESTION 2(B) CONTINUED

Continue your response to Question 2(b) below:

(Concludes on next page →)

Final analytical evaluation & concluding historical verdict for Question 2(b):

• EDEXCEL SECTION A: QUESTION 1 FEATURE DRILLS [4 MARKS • 5 MINS] **2 SEPARATE QUESTIONS • 2 MARKS EACH**

Examiner Mark Scheme: Award 1 mark for identifying a valid historical feature + 1 mark for supporting factual detail. Supporting detail must directly develop the named feature.

1(a) Describe one feature of Harvey Cushing's surgical techniques in operating on head wounds on the Western Front. **[/2]**

FEATURE:

DETAIL:

1(b) Describe one feature of the surgical methods used by Harold Gillies at Queen's Hospital, Sidcup. **[/2]**

FEATURE:

DETAIL:

PUPIL NAME: _____ TARGET: _____ CURRENT GRADE: _____

ENQUIRY	ASSESSMENT COMPONENT FOCUS	PAGE	SCORE	TEACHER COMMENT & NEXT STEPS
KT5.1	**Q1:** Describe Two Features of Trenches / Arras [4m]	p. 7	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(a): Source Utility Assessment (Trench Design) [8m]	p. 5–7	/ 8	
KT5.2	**Q1:** Describe Two Features of Trench Foot / Fever [4m]	p. 11	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(b): 4-Prompt Follow-Up Matrix (Living Conditions) [4m]	p. 9–11	/ 4	
KT5.3	**Q1:** Describe Two Features of Shrapnel / Poison Gas [4m]	p. 15	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(a): Source Utility Assessment (Artillery Trauma) [8m]	p. 13–15	/ 8	
KT5.4	**Q1:** Describe Two Features of Dressing Station / FANY [4m]	p. 19	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(b): 4-Prompt Follow-Up Matrix (Stretcher Transport) [4m]	p. 17–19	/ 4	
KT5.5	**Q1:** Describe Two Features of Thomas Splint / X-Rays [4m]	p. 23	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(a): Source Utility Assessment (Mobile X-Rays) [8m]	p. 21–23	/ 8	
KT5.6	**Q1:** Describe Two Features of Cushing / Gillies Surgery [4m]	p. 27	/ 4	MARKING FEEDBACK • TARGETS:
	Q2(b): 4-Prompt Follow-Up Matrix (Cambrai Blood Bank) [4m]	p. 25–27	/ 4	

• DIGITAL LEARNING & RETRIEVAL HUB • INTERACTIVE LESSONS & QUIZZES

Interactive Textbook • 10/20 Questions Per Enquiry

KT5.1	KT5.2	KT5.3	KT5.4	KT5.5	KT5.6
Trenches & Mud	Trench Illness	Trauma & Gas	Evacuation Chain	Splints & X-Rays	Blood & Plastic
LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB	LESSON HUB
PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ	PRACTICE QUIZ
Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]	Score: [/ 10]

Western Front Spec Complete: 16 marks in the bag, zero gangrene, and not a single complaint to the War Office.